

# Digital Sovereignty for Kashmir

From Metadata Today to Our Own Servers Tomorrow — A Practical Roadmap

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# Tonight's Roadmap

Why This, Why Now

Internet & Electricity: The Foundation

Physical Environment

Starting with VPS: Our On-Ramp

Hardware for Full Scale: 100,000 Users

Software & the Skills We Need

The Hope

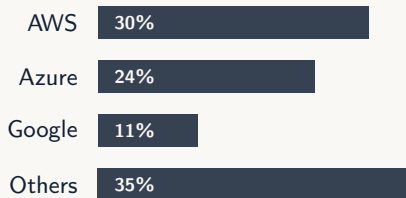
## Why This, Why Now

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# The cost of renting our own knowledge

## Where the world's data actually lives

- Global cloud infrastructure spend is concentrated in three American firms — roughly two-thirds of the market, by public analyst estimates (Synergy Research, Canalys).
- Every one of them is bound by the U.S.  
**CLOUD Act:** a US company must hand over data on request, *regardless of which country the server sits in*.
- “Data residency in India” on an American platform is a comforting label, not sovereignty.



Indicative global cloud infrastructure share, 2025–26 analyst estimates.

The question isn't whether we depend on cloud infrastructure — we already do. The question is *whose* infrastructure, under *whose* law, and who profits.

# OLK today: a working example, and its ceiling

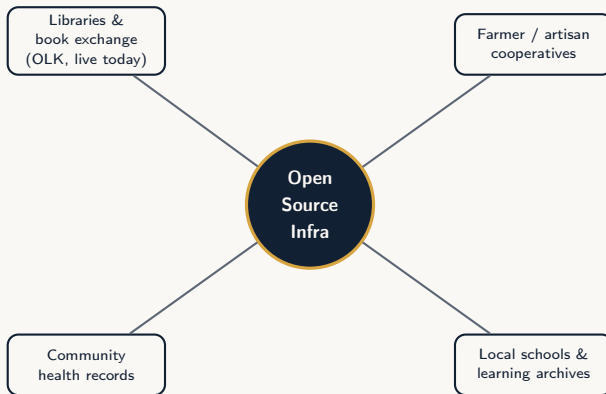
## What we already built

- **Open Library Kashmir** — a privacy-first book donate/lend platform for the region. Real users, real listings, real requests.
- Fully open-source stack: Next.js, React, Tailwind, PostgreSQL via Supabase. Every layer is software we control.
- Runs today on Supabase's hosted infrastructure — physically in AWS's Mumbai region (ap-south-1).

## The ceiling

- Geographically close  $\neq$  jurisdictionally ours. It is still Amazon's rack, Amazon's power, Amazon's law.
- Supabase itself is a US/EU company with its own terms, pricing, and continuity risk.
- **The good news:** because it's 100% open-source, nothing needs to be rewritten to move it. It needs to be *redeployed* — onto infrastructure we own.

## Beyond one app: why this could matter for millions



OLK is the *first* baby step, not the destination. The same self-hosted stack — Postgres, auth, storage, a web front end — is the pattern for every community institution in the region that currently has no digital backbone of its own, from Kashmir to the wider Indian subcontinent.

# Internet & Electricity: The Foundation

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# How much internet do we actually need?

## Sizing the pipe for 100,000 users

Registered users = 100,000

Concurrent (peak)  $\approx 8\% = 8,000$

Avg. active payload  $\approx 30 \text{ KB/s/user}$

Sustained peak =  $8,000 \times 30 \text{ KB/s}$   
 $\approx 240 \text{ MB/s} \approx 2 \text{ Gbps}$

Burst headroom ( $4\times$ )  $\approx 8\text{--}10 \text{ Gbps}$

- A typical home/business fiber line in Srinagar (100 Mbps–1 Gbps) is **an order of magnitude short** of what full scale needs.
- We need **datacenter-grade, redundant** uplinks: multiple carriers, multiple physical paths, not one ISP line.
- Kashmir's history of internet shutdowns means single-path connectivity is not just slow — it is a single point of *total* failure.

**Design principle:** keep a CDN / edge cache *outside* the valley (e.g. a Chinese cloud edge in Hong Kong or Singapore) so read-heavy traffic survives a local shutdown even if writes pause.



# Electricity: the honest picture

## Where J&K's grid stands (2026)

- Billing efficiency improved 56% (FY21-22) → 77% (FY25-26) under the RDSS reform program — real progress, still not hosting-grade.
- KPDCL routinely issues *scheduled* maintenance outages across Srinagar, Pulwama, Baramulla (e.g. 11–18 July 2026).
- Unscheduled outages remain common, especially in winter peak-load months.

## What that means for us

- Grid power = supplementary, not primary, from day one.
- Rack-level **online double-conversion UPS**, sized to full server load, minimum 30–60 min autonomy.
- Diesel generator, auto-transfer switch, 24–72h fuel reserve for extended outages.
- Solar + battery as a long-term supplement (strong insolation Apr–Oct, weak in winter — *never* the sole source).

# Physical Environment

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# The room the servers live in

## Cooling

- Kashmir's cool climate is an asset: free-air / economizer cooling works most of the year.
- Precision AC only needed for summer peak and humidity control.
- Sealed rack + positive-pressure filtration against valley dust and pollen.

## Safety

- Clean-agent fire suppression (FM-200/Novec) — never water sprinklers over live racks.
- Smoke and heat detection wired to auto shutdown.

## Physical security

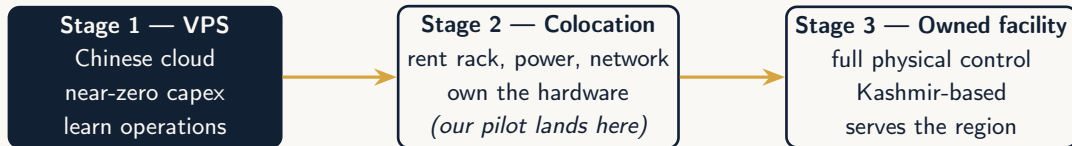
- Single access-controlled room; logged entry, CCTV.
- Locking rack cabinets; no shared building access.
- Humidity/temperature monitoring with SMS/email alerting.

For the **pilot** (slide 29), all of this fits in a single well-prepared room with one rack — not a purpose-built datacenter. Full 100k scale is a later, separate capital project.

## Starting with VPS: Our On-Ramp

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# Why VPS first: the staged path to sovereignty



- **Stage 1** costs almost nothing to start and teaches the team real operational skills — Linux, backups, monitoring — before any capital is at risk.
- Every stage runs the *same* open-source software stack. Moving stages is a migration, not a rewrite.
- We are not choosing between VPS and our own datacenter. We are choosing the *order* in which we build confidence and capability.

# The Chinese cloud landscape

## Why Chinese providers, specifically

- Alibaba Cloud, Tencent Cloud and Huawei Cloud are three of the world's largest hyperscalers by capacity — mature, globally reachable, and **outside** the US CLOUD Act's jurisdiction.
- All three operate international sites (Hong Kong, Singapore) purpose-built for customers outside mainland China — no need to navigate mainland-only compliance.
- Huawei Cloud in particular runs on the *same* CPU and OS family (Kunpeng / openEuler) we would later deploy on our own hardware — one skill set, two stages.
- This is a deliberate, practical choice for this presentation: every option from here on is Chinese-made. No AWS, no Azure, no Google.

# Alibaba Cloud vs. Tencent Cloud vs. Huawei Cloud

|                          | Alibaba Cloud                                          | Tencent Cloud                                         | Huawei Cloud                                                |
|--------------------------|--------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------|
| Best for us              | Broadest footprint outside China; mature intl. billing | Lighthouse = simple all-in-one VPS, ideal first pilot | Kunpeng + openEuler = same stack as our own future hardware |
| Nearby region            | Singapore, Hong Kong                                   | Hong Kong, Singapore                                  | Hong Kong, Singapore                                        |
| Est. latency to Srinagar | ~40–80 ms                                              | ~40–80 ms                                             | ~40–80 ms                                                   |
| Entry product            | ECS (Elastic Compute Service)                          | CVM / Lighthouse                                      | ECS (incl. Kunpeng instances)                               |
| Support in English       | Good                                                   | Good                                                  | Improving                                                   |

**Recommendation:** start on *Tencent Lighthouse* or *Alibaba ECS* for the cheapest, fastest first pilot; plan the eventual on-prem hardware around *Huawei's* Kunpeng/openEuler ecosystem for continuity.

# Getting a Chinese VPS as a foreign, non-mainland user

## Practical steps

1. Register on the **international** site (`intl.alibabacloud.com`, `intl.cloud.tencent.com`, `huaweicloud.com/intl`) — not the mainland site.
2. Choose a **Hong Kong or Singapore** region: this avoids China's mainland **ICP license** requirement, which only applies to sites physically hosted inside mainland China.
3. Identity verification: passport + organisation documents (we would register OLK/the collective formally to do this cleanly).
4. Payment: international credit card; PayPal or Alipay-linked options exist on some products.
5. Start **pay-as-you-go** on a small instance before committing to any reserved/annual plan — validate our workload first.

None of this requires a China-based business entity. It requires an afternoon of paperwork.



**Hardware for Full Scale:  
100,000 Users**

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# Storage 101: what we need, and how much

## SSD vs. NVMe

- SATA SSD: fast, cheap, capped around 550 MB/s by the SATA interface.
- **NVMe**: talks directly over PCIe lanes — multi-GB s<sup>-1</sup> throughput and far higher IOPS. Essential for a database doing thousands of small random reads/writes (exactly our workload).

## Sizing for 100,000 users

- Avg. user footprint (profile + listings + messages)  $\approx 5 \text{ MB} \rightarrow 500 \text{ GB}$
- Book cover images,  $\sim 300\text{k listings} \times 300 \text{ KB} \approx 90 \text{ GB}$
- DB overhead: indexes, WAL, replication, backups (3–5 $\times$ )
- **Design target: 2–5 TB hot NVMe**, plus a separate warm/archive tier for older data and backups

This is exactly why our pilot's **5 TB NVMe** spec (slide 29) isn't arbitrary — it is sized with real headroom for 10–20k users today and a clear growth path toward 100k.

# RAID: redundancy is not optional

| Level          | How it works                 | Fault tolerance | Usable capacity |
|----------------|------------------------------|-----------------|-----------------|
| RAID 0         | Striped, no redundancy       | None            | 100%            |
| RAID 1         | Mirrored pair                | 1 disk          | 50%             |
| RAID 5         | Striped + 1 parity disk      | 1 disk          | $(n - 1)/n$     |
| RAID 6         | Striped + 2 parity disks     | 2 disks         | $(n - 2)/n$     |
| <b>RAID 10</b> | Mirrored pairs, then striped | 1 per mirror    | 50%             |

**Our pick:** RAID 10 for the live database volume — best performance *and* redundancy, worth the capacity cost for the system of record. RAID 6 is acceptable for the bulk image/backup archive, where raw capacity matters more than latency.

# ECC RAM: why a bit-flip matters here

- Consumer RAM has no protection against random bit-flips caused by cosmic rays or electrical noise — rare, but real, and worse at scale (more DIMMs, more hours of uptime).
- **ECC (Error-Correcting Code) RAM** detects and silently corrects single-bit errors in real time.
- For a *system of record* — who owns which book, who borrowed what — a silently corrupted row is far worse than a slow page load. This is not a place to save money.

## Sizing

- PostgreSQL wants RAM close to its working set. **128 GB** gives ~32 GB `shared_buffers` plus generous OS page cache — comfortable for the 10–20k pilot.
- Full 100k scale: 256–512 GB per node, or split the workload across several nodes with a distributed database (slide 24) instead of one ever-bigger box.

# The Chinese hardware stack, end to end

| Layer            | Chinese-made option                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| CPU              | Huawei <b>Kunpeng</b> 920/950 (ARM64; the new TaiShan 950 SuperPoD ships up to 192-core/384-thread, Q1 2026)          |
| Server chassis   | Huawei <b>TaiShan</b> , <b>Inspur</b> , <b>XFusion</b> (H3C) — all ship Kunpeng/openEuler-certified rack servers      |
| Storage          | <b>YMTC</b> / <b>ZhiTai</b> NVMe — domestic 3D NAND (Xtacking 4.0); flagship PCIe 5.0 models reach ~10–15 GB/s reads  |
| Operating system | <b>openEuler</b> — Huawei's open-source Linux, ARM64-native, >50% share of newly deployed China server OS (IDC, 2024) |
| Networking       | Huawei / H3C switches, 10–25 GbE                                                                                      |

Kunpeng-based servers already hold over 20% of the China server market (IDC, 2024) — this is not a fringe or unproven stack, it is mainstream infrastructure we can simply choose.

## A concrete Bill of Materials — 100k-scale node

| Component       | Spec                                          |
|-----------------|-----------------------------------------------|
| Chassis         | 2U TaiShan / XFusion rack server, dual PSU    |
| CPU             | 1–2× Kunpeng 920-class (ARM64)                |
| RAM             | 256 GB ECC RDIMM                              |
| Boot volume     | 2× enterprise SATA SSD, RAID 1                |
| Database volume | 6× ZhiTai/YMTC NVMe, RAID 10 (~5–8 TB usable) |
| Network         | Dual-port 10/25 GbE                           |
| OS              | openEuler LTS                                 |
| Rack support    | UPS (online double-conversion) + PDU          |

Pricing fluctuates with CNY/INR rates and is best obtained as a direct quote from an XFusion/Inspur reseller or a Huawei

Cloud dedicated-host equivalent — treat this as a specification, not a quote.

## Software & the Skills We Need

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# The software stack: what changes, what doesn't

## Stays the same

- **PostgreSQL** — what Supabase runs today, and what OLK already depends on. Open-source, portable, battle-tested.
- Our entire app layer (Next.js, Auth, Storage APIs) is already license-free to redeploy anywhere.

## New, Chinese, and worth knowing

- **openGauss** — Huawei's own PostgreSQL fork, built for distributed, PB-scale workloads.
- **TiDB** (PingCAP) — MySQL-wire-compatible distributed SQL, strong open-source community.
- **OceanBase** (Ant Group) — financial-grade distributed OLTP at extreme concurrency.

Because openGauss is a PostgreSQL fork, our migration path exists but isn't urgent. We only need to reach for a distributed database like this once a single well-specced node (our pilot) stops being enough — likely well past 100k users.



# Supabase self-hosting: we are already sovereignty-ready

- Supabase is a thin, open-source layer over Postgres: PostgREST (API), GoTrue (auth), Realtime, Storage API, Studio — all Apache/MIT licensed, all Docker-composable.
- We already run this *exact* stack locally via the Supabase CLI and Docker for development.
- Self-hosting in production is the same containers, pointed at our own Postgres, on our own machine, behind our own domain.

**The headline point of this whole talk:** we do not need to rewrite Open Library Kashmir to make it sovereign. We need to *redploy* it — onto a VPS first, then our own hardware.

## Skills matrix: what our team must build

| Skill                                                        | Why it's needed                      | Have today?   |
|--------------------------------------------------------------|--------------------------------------|---------------|
| Linux / openEuler administration                             | Every layer runs on it               | Partial       |
| Docker & container orchestration                             | How Supabase's services are deployed | Yes           |
| PostgreSQL DBA (backup, tuning, replication)                 | Data integrity at scale              | Partial       |
| Networking & firewalling (multi-carrier failover)            | Shutdown-resilient uplink            | No — to learn |
| Security (patching, encryption at rest, intrusion detection) | System of record for real people     | No — to learn |
| Facilities basics (power, cooling)                           | Slides 9 & 11                        | No — to learn |

This is a call for volunteers as much as it is a call for hardware budget. Every gap in the “No” column is a role someone in this room could own.

## The 3-2-1 rule, adapted for us

- 3 copies of the data — production + two backups.
- 2 different media/locations — e.g. local NVMe snapshot + a separate disk or machine.
- 1 copy offsite — a different city in J&K, or an **encrypted** bucket on Alibaba OSS / Tencent COS, used strictly as cold backup storage, never as the primary host.

Everything is encrypted *before* it leaves our building. An offsite Chinese cloud bucket used purely for encrypted backup carries none of the sovereignty compromise of hosting live traffic there.

# The Hope

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# The proposal: one machine, real capacity, right now

## The pilot

- **10,000–20,000 users**, served from a *single* well-specced machine.
- **5 TB usable NVMe (RAID 10) + 128 GB ECC RAM** + Kunpeng-class CPU, openEuler, PostgreSQL + self-hosted Supabase.
- Chinese-made hardware and hosting throughout — no Western dependency, from the VM to the silicon.

## Why this scope

- Small enough to fund and operate with volunteer skills we can build this year.
- Large enough to be a *real* platform, not a toy demo.
- Proves every claim in this talk before we ask for datacenter-scale capital.

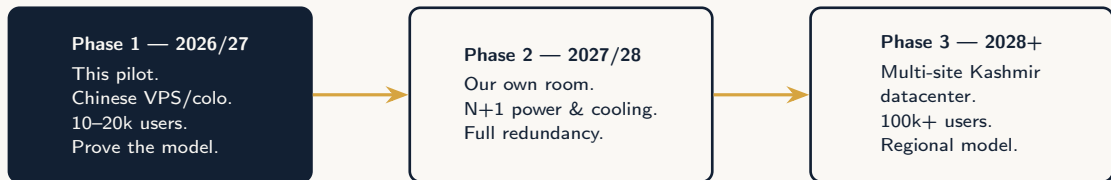
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Headroom     $20,000 \text{ users} \times 5 \text{ MB} \approx 100 \text{ GB active data} \rightarrow 5 \text{ TB} / 100 \text{ GB} \approx 50\times \text{ headroom}$

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Enough for years of growth, every book cover, every backup snapshot, without touching the hardware again.

# The road from here



**What we're asking for tonight:** a funding commitment for the Phase 1 bill of materials, volunteers against the skills matrix, and a room to eventually call our own. This is buildable in months, not decades — and it starts the moment we agree to start.

Our data. Our servers. Our region. **Let's begin.**